UNIVERSIDAD DE PAMPLONA

Demostraciones de la Distribución Weibull

Materia: Estadística Avanzada

Docente: Diego Enrique Pico Lache

Estudiante: Juan Carlos Pedraza Villamizar

Programa: Matemática Aplicada

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Distribución Weibull

Def: Se dice que la V.A. $X \sim W(\alpha, \beta)$ de parámetros α, β si su función de densidad dada por:

$$f_X(x) = \begin{cases} \alpha\beta x^{\beta-1} e^{-\alpha x^{\beta}} & , \quad x \geq 0 \\ 0 & , \quad x < 0 \end{cases}$$

$X \sim W(\alpha, \beta)$ indica que la variable aleatoria X tiene distribución Weibull de parámetros α, β.

$$\star\ E[X] = \left(\frac{1}{\alpha}\right)^{\frac{1}{\beta}} \Gamma\left(1 + \frac{1}{\beta}\right)$$

$$\star\ \text{Var}[X] = \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} \left[\Gamma\left(1 + \frac{2}{\beta}\right) - \Gamma^2\left(1 + \frac{1}{\beta}\right)\right]$$

Demostración que es una función de densidad

$$\star \int_0^{\infty} f_X(x)\, dx = 1$$

$$\star \int_0^{\infty} \alpha\beta x^{\beta-1} e^{-\alpha x^{\beta}}\, dx \qquad \begin{array}{l} \star\ u = \alpha x^{\beta} \\ \star\ du = \alpha\beta x^{\beta-1}\, dx \end{array}$$

$$\Rightarrow \int_0^{\infty} e^{-u}\, du = \lim_{b\to\infty} \left[-e^{-u}\right]_0^b$$

$$\Rightarrow \cdots = -\left[\lim_{b\to\infty} e^{-b} - \lim_{b\to\infty} e^{-0}\right]$$

$$\Rightarrow \cdots = -[0 - (1)]$$

$$\boxed{\int_0^{\infty} \alpha\beta x^{\beta-1} e^{-\alpha x^{\beta}}\, dx = 1}$$

Demostración $E[X]$

$$E[X] = \int_0^\infty x \left[\alpha\beta x^{\beta-1} e^{-\alpha x^\beta}\right] dx$$

$$\star\ u = \alpha x^\beta \qquad \star\ x = \left(\frac{u}{\alpha}\right)^{\frac{1}{\beta}}$$
$$\star\ du = \alpha\beta x^{\beta-1}\, dx$$

$$\begin{aligned}
\Rightarrow \cdots &= \int_0^\infty \left(\frac{u}{\alpha}\right)^{\frac{1}{\beta}} e^{-u}\, du \\
&= \int_0^\infty \left(\frac{1}{\alpha}\right)^{\frac{1}{\beta}} u^{\frac{1}{\beta}} e^{-u}\, du \\
&= \left(\frac{1}{\alpha}\right)^{\frac{1}{\beta}} \int_0^\infty u^{\frac{1}{\beta}} e^{-u}\, du
\end{aligned}$$

$$\boxed{\therefore E[X] = \left(\frac{1}{\alpha}\right)^{\frac{1}{\beta}} \Gamma\left(1 + \frac{1}{\beta}\right)}$$

Demostración $E[X^2]$

$$E[X^2] = \int_0^\infty x^2 \left[\alpha\beta x^{\beta-1} e^{-\alpha x^\beta}\right] dx$$

$$\star\ u = \alpha x^\beta \qquad \bullet\ \left(\frac{u}{\alpha}\right)^{\frac{2}{\beta}} = x^2$$
$$\star\ du = \alpha\beta x^{\beta-1}\, dx$$

$$\begin{aligned}
\Rightarrow \cdots &= \int_0^\infty \left(\frac{u}{\alpha}\right)^{\frac{2}{\beta}} e^{-u}\, du \\
&= \int_0^\infty \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} u^{\frac{2}{\beta}} e^{-u}\, du \\
&= \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} \int_0^\infty u^{\frac{2}{\beta}} e^{-u}\, du
\end{aligned}$$

$$\boxed{\therefore E[X^2] = \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} \Gamma\left(1+\frac{2}{\beta}\right)}$$

Demostración $\mathbf{Var}[X]$

$$\begin{aligned}
\mathrm{Var}[X] &= E[X^2] - (E[X])^2 \\
\mathrm{Var}[X] &= \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} \Gamma\left(1+\frac{2}{\beta}\right) - \left[\left(\frac{1}{\alpha}\right)^{\frac{1}{\beta}} \Gamma\left(1+\frac{1}{\beta}\right)\right]^2 \\
\mathrm{Var}[X] &= \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} \Gamma\left(1+\frac{2}{\beta}\right) - \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} \Gamma^2\left(1+\frac{1}{\beta}\right)
\end{aligned}$$

$$\boxed{\mathrm{Var}[X] = \left(\frac{1}{\alpha}\right)^{\frac{2}{\beta}} \left[\Gamma\left(1+\frac{2}{\beta}\right) - \Gamma^2\left(1+\frac{1}{\beta}\right)\right]}$$